

A 10-MINUTE BRIEFING

From ~~Black-Box~~ to *Open*, Auditable AI.

Six ideas. One conclusion. Why the next phase of AI belongs to open weights, small models, and auditable operations.

RISK → RISE → SMALL → MITIGATION → FUTURE

WHAT WE

01 

02 

03 

04 

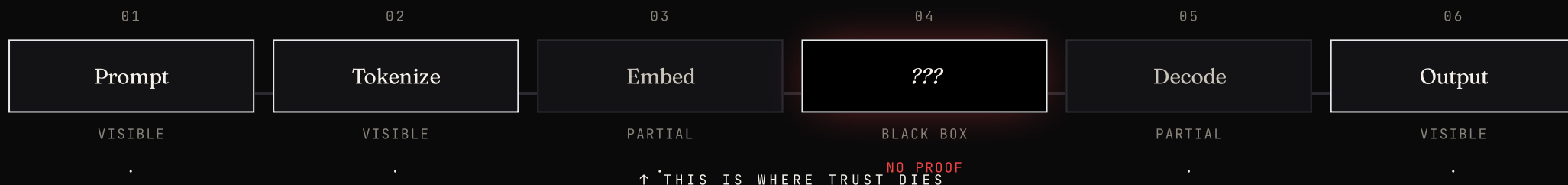
~ 90 SECONDS PER SLIDE • QUESTIONS AT THE END

LET'S BEGIN →

Inside the transformer, the token journey goes *dark*.

WHAT ENTERS AS LANGUAGE EXITS
AS A GUESS — WITH NO RECEIPT

THE JOURNEY OF A TOKEN



No explainability

Attention ≠ business reasoning. The path from token to answer is opaque.



Hallucination

Confident, fluent — and often wrong. Closed APIs offer no source proof.



Data leakage

Prompts, docs, PII flow to a vendor with no audit of training reuse.



Silent model drift

Provider swaps weights overnight. Yesterday's controls no longer apply.



Lock-in & cost shock

Pricing, latency, and access change without warning. No fallback.

"Open" is a spectrum.
The world is racing *right* across it.

TIER	WEIGHTS	ARCH	TRAIN DATA	FINE-TUNE	SELF-HOST
Closed API e.g. GPT, Claude	×	×	×	×	×
Open Weight e.g. Llama, Mistral	✓	✓	×	✓	✓
Open Source e.g. OLMo, Pythia	✓	✓	✓	✓	✓

OPEN-WEIGHT ≠ OPEN-SOURCE. AUDIT TEAMS MUST READ THE LICENSE, NOT THE HEADLINE.

WHY IT'S RISING — FAST

- 70%+
of new enterprise pilots now test open-weight models alongside frontier APIs
- 10×
cheaper inference on a fine-tuned 7B vs. a closed frontier API at scale
- 0
data leaves your VPC when models run on-prem or in a private cloud

Plus: *sovereignty, customization, no vendor lock-in*, and a thriving ecosystem of fine-tunes.

Open weights + small models + *agentic loop* = the new enterprise stack.

OPEN-SOURCE MODELS

Llama 4 META	17B-400B (MoE) Multimodal · long-context · reasoning
Mixtral 8x22B MISTRAL	141B / 39B active MoE reasoning · efficient inference
Qwen3 ALIBABA	0.6B-235B Multilingual · coding · agents
DeepSeek-V3 DEEPSEEK	671B / 37B active Frontier reasoning · open MoE
Gemma 3 GOOGLE	1B-27B Lightweight · multimodal
Nemotron 3 Super NVIDIA	120B / 12B active Agentic · enterprise RAG
Command R+ COHERE	104B Tool-use · enterprise RAG
OLMo 2 AI2	1B-32B Fully open (data + weights)

THE FOUNDATION

SMALL LANGUAGE MODELS

Phi-4 Complex reasoning at small scale	14B
Phi-3.5 mini Edge reasoning · on-device	3.8B
Mistral Small 3.1 Function calling · 128K context · vision	24B
Qwen3 Routing · extraction · multilingual	0.6 / 1.7 / 4 / 8B
Llama 3.2 Mobile · summarization · classification	1B / 3B
Gemma 3 On-device · multilingual	1B / 4B
SmolLM2 Embedded · ultra-low latency	135M - 1.7B
Mistral 7B General-purpose workhorse	7B

THE WORKHORSES

NEMO AGENT TOOLKIT

 Perceive <i>Sense the world</i> 01 > Multi-modal input (text · vision · audio) > RAG over enterprise data > Tool & MCP server discovery
 Think <i>Plan & reason</i> 02 > Cross-framework orchestration > Task decomposition & routing > Cost-aware model selection
 Act <i>Execute safely</i> 03 > Tool calls via MCP > Guardrails · least-privilege scopes > Prompt-injection & jailbreak shields
 Learn <i>Improve continuously</i> 04 > Evaluation & golden datasets > Cost · latency · accuracy tracking > Security testing & tool-poison checks

THE LOOP

CHEAPER · FASTER · PRIVATE · FINE-TUNABLE

Agentic + SLMs = 10× cheaper, on-prem, auditable digital workers — that *perceive, think, act, and learn* without leaking your data.

Open-source doesn't just *cut cost*. It closes the *trust gap*.

EACH BLACK-BOX RISK → AN OPEN-SOURCE ANSWER

No explainability	→	Inspect weights, attention, and reasoning traces — locally.
Hallucination	→	Ground every answer in your own RAG over your own data.
Data leakage	→	Run on-prem or in your VPC. Nothing leaves.
Silent model drift	→	Pin the exact weight hash. Upgrade only on your schedule.
Lock-in & cost shock	→	Swap providers and hardware without a rewrite.

WHAT COMES NEXT

01 Hybrid by default

Open-weight SLMs handle 80% of calls. Frontier APIs for the long tail.

02 Agentic + auditable

Every tool call logged, every model version pinned, every action replayable.

03 On-device intelligence

Phones, laptops, factories. Inference moves to where the data lives.

04 Open beats closed

Linux won servers. Open weights will win the AI layer — for the same reasons.

The winners will run a *portfolio* — not a single vendor.

One picture. Two worlds. *Pick yours.*



trust the vendor



OPAQUE



THEIR CLOUD



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DRIFT

~~Can't audit weights~~
~~Can't pin versions~~
~~Can't run offline~~
~~Can't avoid lock-in~~

OUTCOME

Powerful. Expensive. Unaccountable.



trust the evidence



TRANSPARENT



YOUR EDGE



10× CHEAPER



PINNED

✓ Audit every weight
✓ Pin every version
✓ Run on-prem · on-device
✓ Switch vendor anytime

OUTCOME

Powerful. *Affordable.* Accountable.

10×

LOWER INFERENCE COST

100%

DATA SOVEREIGNTY

0

VENDOR LOCK-IN

∞

AUDIT REPLAY

From ~~black-box guesswork~~ to *open, auditable intelligence*. That's the shift.

THANK YOU
QUESTIONS WELCOME